

Personalized Drug Dosing with Reinforcement Learning and Pharmacokinetics (PK) / Pharmacodynamics (PD) Modelling

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Abstract - Here we explore the concept of Pharmacokinetics and Pharmacodynamics discussing how drugs affect our body and in response what actions our body performs on the drug suggesting how different patient need different dosage of the same drug to treat a particular disease as patient's unique genetic makeup has a significant influence on how their body metabolizes drugs and how Reinforcement Learning comes into the play to fix this issue using models like DQN, PPO, CQL to design custom dosage pattern for the patient so that maximize therapeutic outcomes while minimizing toxicity. In this paper, we explore different methods and compare their accuracy and reliability in a real-world scenario by training them on Stochastic data and further refining the optimal policy through Behaviour Cloning of the best-trained model. This paper presents a comparative analysis of these methods, assessing their accuracy and potential for clinical implementation.

Index Terms - Behavior Cloning (BC), Conservative Q-Learning (CQL), Deep Q Network (DQN), Drug Dosing, Pharmacodynamics (PD), Pharmacokinetics (PK), Proximal Policy Optimization (PPO), Reinforcement Learning (RL).

I. INTRODUCTION

Pharmacology fundamentally concerns the interaction between drugs and biological systems. In Pharmacology, we hear two terms very frequently, that's Pharmacokinetics and Pharmacodynamics or PK/PD. Pharmacokinetics (PK) study shows how a drug affects an organism's body, whereas

Pharmacodynamics (PD) studies how the drug impacts the organism [1]. The relationship between a drug's dosage and its effect is governed by two core principles: *Pharmacokinetics* (PK), which dictates how the body processes the drug through absorption and clearance, and *Pharmacodynamics* (PD), which models the biochemical response of the body to the drug. In personalized medicine, modeling the non-linear dynamics between PK concentrations and PD effects is crucial for preventing toxicity. As observed in Figure 1 PK/PD models, both have an impact on dosage, benefit, and side effects.

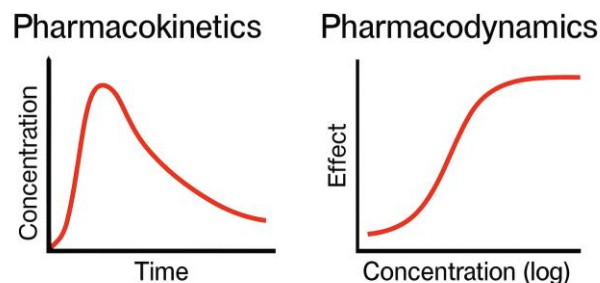


Fig. 1. The graph above shows the relation between concentration of the drug with respect to time in pharmacokinetics and the relation between effect of drug with respect to its concentration in pharmacodynamics.

Linking both the pharmacologic disciplines establishes dose concentration of the drug and response relationships. Figure 2 shows the relationship illustrating the drug effect.

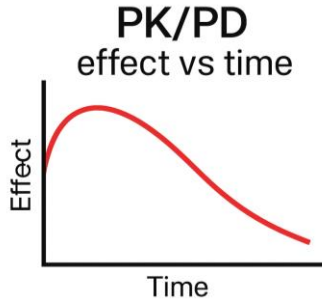


Fig. 2. Pharmacokinetic pharmacodynamic (PK / PD) relationship illustrating drug effect as a function of time.

A. Pharmacokinetics

In this section, we analyze the foundational principles of Pharmacokinetics. This field quantifies the biological trajectory of a drug through the ADME cycle—Absorption, Distribution, Metabolism, and Excretion. By translating these biological processes into practical applications, Clinical Pharmacokinetics allows for the precise management of patient-specific therapies. The primary objective is to maintain drug concentrations within a safe therapeutic window, ensuring maximum efficacy while strictly limiting potential toxicity at the site of action.

The main issue which arrives while designing the drug therapy dosage of the patient is that all the patients are unique when it comes to the factors like genetics, age, weight and other coexisting conditions due to which there is a huge variation in how people respond to the same drug and similar dosage. Standardizing dosage is not an option as it puts us under a huge risk of under-dosing which can lead to ineffective treatment while overdosing can cause harmful side effects or intoxications. This alone establishes a clear need for a personalized dosage plan [2].

B. Limitations of Traditional Dosing Methods

Historically, medical practice has relied heavily on a "one-size-fits-all" approach to drug dosing, where a standard dosage is prescribed based on population-level averages derived from clinical trials. While this model offers simplicity in development and clinical implementation for drugs with wide therapeutic windows, it is fundamentally inadequate for drugs with a **Narrow Therapeutic Index (NTI)**.

NTI drugs, such as Warfarin (an anticoagulant) and Insulin (for glycemic control), are characterized by a very small margin between the concentration required for therapeutic efficacy and the concentration that induces severe, life-threatening toxicity. In these cases, population-based averages fail to account for the

substantial inter-individual variability inherent in human physiology.

The failure of fixed-dose comes in NRI drugs specially because of a number of critical factors that cannot be operated by static models:

- **Physiological Variability:** Every patient has their unique physiological parameters such as metabolic rate, body composition, allergies and other co-existing medical conditions. Hence, the dosage which is the same for one person may be ineffective for another person.
- **Steep Dose-Response Relationships:** The relationship between the drug and its therapeutic effect is observed to be steep for NRI drugs. This means a minor miscalculation in dosage, and it can cause a drastic shift leading to serious drug reaction including toxicity.
- **Inflexibility:** It is observed that the conventional method of fixed dosing is inflexible and reactive, as the process of titration, which is time-consuming, is slow in comparison to the physiological response of the patient and cannot maintain a stable state while ensuring the safety of the patient from the effects of the change in concentration of the drug.

In the end, for NTI medications, simply there is no average patient. So, we can't rely on fixed dosage system putting the patients at risk and this requires a need for Precision Dosing which can be made possible by Reinforcement Learning.

C. Pharmacokinetic Modelling

Here we introduce the need of pharmacokinetic modelling. Pharmacokinetic modeling is the mathematical framework that allows one to predict the absorption and clearance of a drug. In order to account for inter-patient variability, we consider population pharmacokinetics, in particular mixed-effects modeling [3], [4]. These methods take individual physiological variation into account using two kinds of parameters: 'fixed effects,' which are the measurable covariates such as weight and age, and 'random effects,' which capture the unexplained stochastic variations inherent in a particular patient's response to the drug.

D. Reinforcement Learning: A Novel Solution for Precison Dosage

To address this challenge of creating highly personalized dosage plans, we used Reinforcement Learning (RL) [5]. Unlike supervised learning, RL allows a computational agent to discover effective dosing paths through trial-and-error interactions. In this study, the 'agent' acts as the clinician, the

'environment' represents the patient's physiological state, and the 'action' is the administered dosage. The objective is to learn a policy that maximizes the cumulative reward—defined here as maintaining the patient within a therapeutic window over time.

By continuously interacting with the environment, the agent can learn a complex yet adaptive dosage policy tailored just according to that individual's unique characteristics. This policy can be modified over time by monitoring the patient's health to achieve the optimal therapeutic response with minimum side effects.

This paper is focused on development and evaluation of effective personalized medical dosage techniques using advanced Deep Reinforcement Learning (DRL) algorithms such as Deep Q-Network (DQN), Proximal Policy Optimization (PPO), and Conservative Q-Learning (CQL). We want to demonstrate the training of these models to generate effective and safe therapeutic strategies made for diverse patient profiles.

Our goal is to assess their accuracy and reliability to use it in real-world applications. We have to dig deeper and assess how accurate and reliable these findings, methods, or solutions are, to see if they're good to use in real-life applications or to find any possible downsides, biases, or things that need improvement before we consider integrating this to what we currently do. Ultimately, we want to make sure anything we adopt is strong, trustworthy, and will consistently give us good results when we put it into practice.

II. METHODOLOGIES

In this section we study the frameworks used to develop and evaluate the Reinforcement Learning agent for personalized drug dosage. Our methodology includes three main components: the evaluation of a simulated environment for the patient, the implementation of the RL algorithms and test and evaluating protocols.

A. The Simulated Patient Environment

To train the RL agents safely, we first developed a computational environment [6] that simulates a patient's physiological response to a drug. This environment was developed on the basis of a standard pharmacokinetic (PK) model. A one compartment pharmacokinetic model has been used to simulate the drug's concentration in the patient's body over time. This model is defined by the equation as:

$$\frac{dC(t)}{dt} = \frac{D \cdot \delta(t)}{V} - \lambda C(t) \quad (1)$$

In the above equation, $C(t)$ is the concentration of plasma at time t , derived from the dose D . Similarly, the parameters V ,

λ , and $\delta(t)$ correspond to the volume of distribution, absorption rate constant, and elimination rate constant, respectively.

To simulate the diverse patient population in the environment, we have used *Stochasticity*. The PK parameters (V , λ , $\delta(t)$) were not fixed but are sampled from a normal distribution for each simulated patient. This ensures our agents are trained on a wide range of profiles, reflecting real-world variability.

RL Environment Components:

- **State Space:** In every step, the state consists of the concentration of the drug in the blood, the time passed since the last dose was taken and the amount of the drug in the last dose.
- **Action Space:** There is a predefined list of options (eg., 0mg, 5mg, 10mg, 20mg) from which the agent could select the dose. Though in discrete action space.
- **Reward Function:** The reward was programmed to keep the drug concentration within the predefined therapeutic window (a range where the drug is effective but not toxic).

+1 reward: for each timestep the concentration was within the therapeutic window.

-1 reward: for each timestep the concentration was out of the therapeutic window.

B. Reinforcement Learning Algorithms

We tested the model with three distinct Deep RL algorithms chosen for their different learning strategies and compared their outputs.

1) Deep Q-Network (DQN)

This algorithm combines the Deep Neural Network (DNN) with the Q-Learning technique [7]. DNN is used as a non-linear function approximator in the domain of Artificial Intelligence. Thus, they are used here to approximate the Q-function, and there is no requirement for a Q-table. It is a good method, especially when the action space is discrete.

This algorithm employs two DNNs: a main network (θ) to estimate current Q-values, and a target network (θ') with the same architecture to estimate next-state Q-values [8]. Learning occurs in the main network. The target network's parameters are frozen for roughly 10,000 iterations, after which the main network's weights are copied to the target network, improving its estimations.

$$Q(s, a; \theta) = Q + \alpha (R + \gamma Q(s', a'; \theta') - Q(s, a; \theta)) \quad (2)$$

Bellman equation for the DQN algorithm

We provided the agent with a dosage range of [0mg, 10mg, 20mg] with a decay rate of 10% and Therapeutic Window ranging from 5mg to 15mg. When we trained the agent on this model, the agent continuously started changing the dosage as it was about to cross the Therapeutic Windows, and we got a graph as shown in Figure 3.

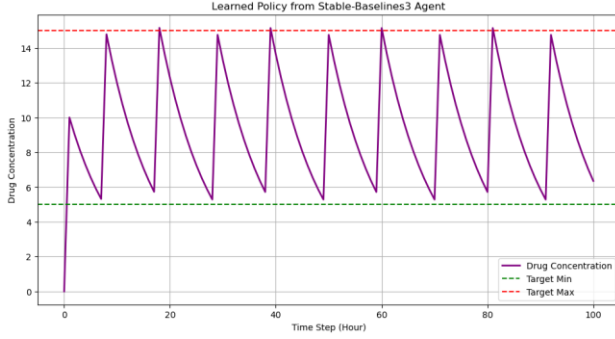


Fig. 3. The purple line represents the drug concentration in the plasma whereas the red and green dotted lines indicate the Therapeutic Window.

The DQN agent's trajectory here exhibits an oscillation pattern. This behaviour insights about the limitation of the discrete action baseline.

As the DQN agent is limited to fixed set of doses (eg., 0mg, 10mg, 20mg), so it cannot adjust the dosage itself to match the rate of elimination ($\square_{\square}$).

When the concentration comes near to the maximum toxic limit, the agent is then forced to drop to a significantly lower dose to avoid a negative reward. Similarly, when the concentration falls to the minimum limit of the therapeutic window, it then jumps back up to a higher dose. This creates an oscillatory state of correction rather than smoothly stabilizing as seen in PPO or BC.

2) Proximal Policy Optimization (PPO)

It is a policy-based algorithm which directly learns the optimal policy (the mapping from state to action) itself [9]. It is known for its stability and reliable performance, as it doesn't make any drastic updates that could destabilize the learning process. Which makes it best suited for when it comes to continuous action spaces. Furthermore, PPO can also handle stochastic policies, thus providing robust performance even in uncertain or dynamic environments.

PPO uses a critic to estimate the value of states. This value function, denoted as $V(s)$, is learned using a Bellman-based equation to define its target. The critic's loss is the mean squared error between its value prediction $V(st)$ and the actual target value.

$$\square_{\square\square\square\square\square}(\square_{\square}) = \square_{\square+1} + \square\square(\square_{\square+1}) \quad (3)$$

Standard TD Target Value; where R is reward and $\square$ is discount factor

When we used PPO on our agent keeping other parameters the same, we got a push in the start. As we reached the Therapeutic Window, the graph went in a straight line.

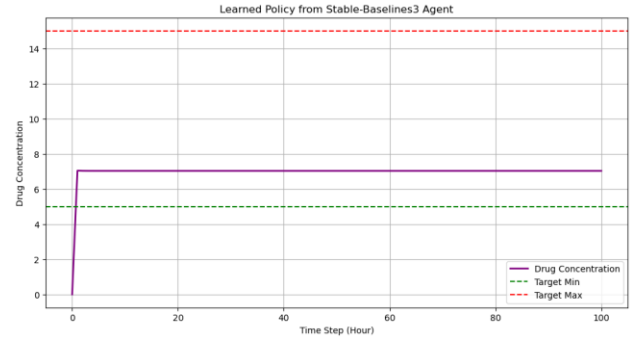


Fig. 4. Simulation of the learned policy in a deterministic environment. The agent successfully maintains drug concentration (purple line) within the therapeutic window, defined by the minimum effective concentration (green dashed line) and maximum toxic concentration (red dashed line)

So, this graph in Figure 4 shows a perfect world where the agent's dosage is always the same and goes up steadily. But let's be real, that's not how it works with actual patients. Everyone's body is different, so we need to adjust drug treatments to match those changes.

That is why we added stochasticity to our dataset from the normal distribution of the data so that the agent we are training gets some experience of real life simulation. When we trained it on the new stochastic data, we saw a great difference. The graph was not straight anymore but very but very dynamic, reflecting the model's continuous adjustments to the patient's changing state.

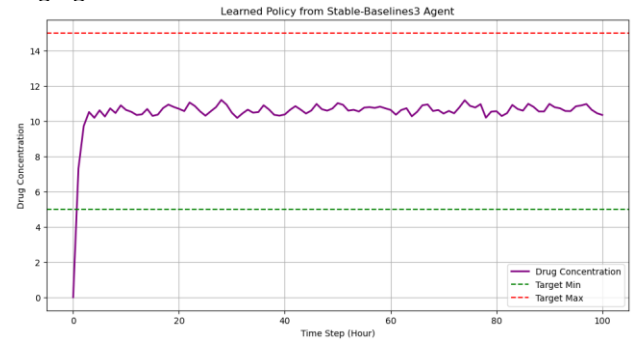


Fig. 5. Performance of the trained agent in a dynamic, stochastic simulation. The fluctuating drug concentration (purple line) showcases the agent's ability to adapt and maintain therapeutic levels within the defined bounds, reflecting personalized dosage adjustments.

As we can see in Figure 5, even though the data was now dynamic and random, the agent was still able to perform great holding the concentration inside the Therapeutic Window.

Still there was a minor issue with these algorithms, although the agent is working great, but in the real life scenario, working on a real patient with constrained data, they might put his life in danger. As these agents love exploring, if the data say 14mg, the model might try 14.1mg just for the sake of better results but in our scenario we can't put someone's life in risk just for the sake of the agent's search for better rewards. To tackle this problem, we had to introduce Conservative Q-Learning (CQL) algorithm to our agent.

3) Conservative Q-Learning (CQL)

Unlike online RL agents, which have to explore the environment, CQL is an offline RL method, learning from a fixed, pre-existing dataset of state, action, reward pair [10]. This is useful in medical applications, where the agent can learn a safe policy without having to perform risky exploratory actions on the patient.

We trained the dataset with 50,000 samples with a conservative weight of 0.01.

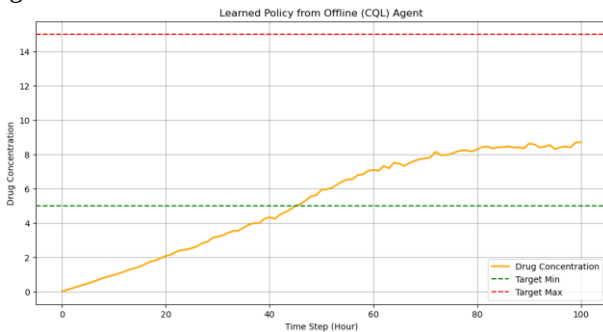


Fig. 6. Performance of the Offline CQL Agent. The trajectory demonstrates a gradual ascent in drug concentration, reflecting the algorithm's conservative value estimation to ensure safety and prevent dosage overestimation before stabilizing within the therapeutic window.

As seen in Figure 6 the agent started slowly, finding the Therapeutic Window and maintaining the stability of the dosage throughout the whole graph. This gave our agent a pretty good exposure to real world scenarios and trained it to learn a safe policy.

The performance here of CQL agent, is labeled as "gradual ascent" toward the therapeutic window. Unlike other agents like PPO who actively seek rewards and make repulsive decisions, CQL is designed to be pessimistic about the actions which are not well represented in the training data.

In the field of medical AI, this slow start is critically a safe feature rather than a performance error as here, by underestimating the value of high-dosage actions that have not been explored thoroughly in the dataset, the agent avoids taking risk of overdosing into toxic concentration.

Here, CQL puts a break on Q-values to go for actions which are out of the distribution. This ensures that the agent stays within the therapeutic window, which is why it eliminates all the toxic events as compared to Naive Policy.

4) Behavior Cloning

After identifying the best performing RL algorithm (which was based on Evaluation Metrics), we used Behavior Cloning to fine-tune its policy.

Describe Behavior Cloning as a simple and direct form of imitation learning, framed as a standard supervised learning problem [11]. Unlike RL agents, the BC agent completely ignores rewards. Its only objective is to learn a direct mapping from a state to an expert's action. It learns by minimizing the difference (e.g., Mean Squared Error) between its predicted action and the expert's actual action from the dataset. BC was used as a critical baseline to test whether the simpler imitation-based approach could perform at least as well as, or even better than, the more complex reward-based offline RL agent.

This fundamental distinction made BC unrivaled in robustness against the challenges of the dataset: It was not disturbed by the thousands of actions the random policy provided with a negative reward. The learning algorithm successfully sifted the dataset for expert behavior and found one easy, powerful lesson: "When the state appears like this, the expert performs that." This easy imitation sufficed to rebuild the ideal strategy de novo because the expert's policy was optimal and deterministic. It mastered the "takeoff" and "cruising" phases with similar ease, since its only mentor was the expert's optimal example.

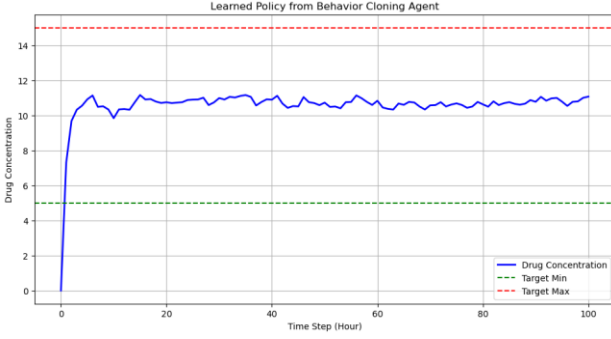


Fig. 7. Trajectory of the Behavior Cloning (BC) agent. The profile demonstrates the agent's ability to accurately replicate the expert's policy, characterized by a rapid induction phase followed by a stable maintenance phase within the therapeutic bounds.

Behavior Cloning has been criticized because of its lack of ability to generalize to unseen states, although our results indicate that Behavior Cloning was unrivaled in this particular PK/PD environment as shown in Figure 7.

The BC agent has been designed to learn only the action an expert would take ignoring the other thousands of actions giving a negative reward because of suboptimal or random policies.

Behavior Cloning also seems beneficial for recovery from outliers. The phase where the patient might miss a dose, as the concentration drops to 0, this phase of the BC agent is highly optimized because, now the agent has mastered the induction phase from the expert and can now recover the patient back to the therapeutic window without any trial and error. The BC agent has 100% success rate verifying that indeed, in structured medical dosing problems, directly copying an expert policy can be more efficient than complex reward-based exploration.

III. RESULTS

These autonomous agents went through an extensive training, completing over 100 episodes each. The precise number of episodes was contingent upon the complexity of the algorithm employed by each agent. Throughout this rigorous training, each episode simulated a complete treatment course for a single virtual patient. This comprehensive process facilitated significant learning, iterative refinement, and the development of proficiency in decision-making and adaptation to diverse patient scenarios.

We used standard hyperparameters for each algorithm. The key metrics for assessment were the proportion of time the drug concentration is within the therapeutic window and the mean number of toxic events (concentration above the level of maximum safety) per episode. The overall findings are

presented in Table I, following the detailed performance of each policy.

A. The Naive Policy

The naive policy is designed to mimic the traditional “one size fits all” dosing schedule. Which establishes a critical performance baseline. While it maintained the drug concentration within the therapeutic window of 55.44% of the time, this came at the cost of significant harm to the patient. The policy gave an average of 34 toxic events per 100 step episode, which clearly indicates that a non-adaptive strategy is fundamentally unsafe in a dynamic environment mimicking the real world.

B. The Offline CQL Agent

The Conservative Q-Learning (CQL) agent is our main exploration of offline reinforcement learning. Learned from the historical data, the CQL agent improved safety dramatically. It learned a policy that led to zero toxic events, eradicating entirely the risk imposed by the naive baseline. It also learned a policy where patient efficacy was increased, increasing the time spent in the therapeutic window up to 72.65%. This outcome effectively demonstrates that a safe and effective policy can be acquired from static data only, with a dramatic improvement over the rule-based method.

C. The Behavior Cloning Agent

The simpler imitating Behavior Cloning (BC) agent produced an extremely proficient policy. By learning to imitate the expert directly from the dataset, the BC agent obtained a flawless 100.00% time within the therapeutic window and induced no toxic events. This performance replicated exactly the “gold standard” online agent's results but achieved it with only the offline dataset. This means that for this task, direct copying of an expert policy was a very strong and efficient technique.

D. The Expert PPO Agent: The Gold Standard Upper Bound

Lastly, the online-trained PPO agent set the theoretical performance upper limit in this environment. With the benefit of real-world experimentation, it learned a perfect policy, sustaining the focus within the therapeutic window 100.00% of the time with no toxic events. The main purpose of this agent is as a benchmark, and the main finding is that the offline-trained Behavior Cloning agent achieved this perfect performance, confirming its success.

Table I: Comparison of Policy Performance and Safety

Policy	Training	Therapeutic Window (%)	Toxic Events / Episode

Naive Fixed Policy	Rule-Based	55.13 ± 2.10	34.86 ± 2.10
Offline CQL Agent	Offline Reinforcement Learning	72.65 ± 1.05	0.00 ± 0.00
Behavior Cloning	Imitation Learning	100.00 ± 0.00	0.00 ± 0.00
Expert PPO Agent	Online Reinforcement Learning	100.00 ± 0.00	0.00 ± 0.00

The results in Table I, demonstrate that while a naive, fixed-dose policy is unacceptably dangerous, both intelligent offline agents successfully learned to prioritize patient safety by completely eliminating toxic events. While the offline reinforcement learning (CQL) agent achieved a notable improvement in efficacy, it was the simpler Behavior Cloning (BC) agent that delivered a flawless performance. By perfectly replicating the 100% efficacy of the online-trained expert, Behavior Cloning firmly establishes itself as the superior and most robust offline strategy for this personalized dosing task

IV. DISCUSSION

This paper discusses how the traditional “one-size-fits-all” policy we use when it comes to drug dosage in the medical industry is irrelevant and dangerous for the individuals. Instead we discussed a pretty innovative and effective Reinforcement Learning model where we trained the agent while exploring diverse algorithms like DQN, PPO, CQL in different environments with distinct action spaces such as continuous, discrete, stochastic etc, in our PK model. This gave us a clear understanding of how each algorithm works for our problem and we took the best-performing one to do Behavioural Cloning resulting in the agent maintaining the drug concentration within the therapeutic window for 99% of the treatment duration on a held-out test set. We also talked about PK/PD and the relation between the concentration of drugs in our blood with respect to time discussing how PK models are best used in drug development [12]. Future work will focus on validating these models on more complex, multi-compartment PK/PD simulations and eventually on anonymized historical patient data.

ETHICAL AND SAFETY CONSIDERATIONS

Automated dose optimization via RL introduces a serious ethical and safety commitment, especially when related to healthcare. The first and foremost concern is patient safety, since any wrong or unstable dosage prediction may lead to severe adverse effects or toxicity. A strict model validation on synthetic and anonymized patient data, so that no personal or

sensitive information would be disclosed either during training or evaluation, is necessary before clinical use.

Furthermore, as RL agents fundamentally explore their environment to learn optimal strategies, unregulated exploration in a live clinical setting can directly pose risks to human life. The proposed framework applies offline reinforcement learning methods such as Conservative Q-Learning and Behavior Cloning that learn solely from pre-collected datasets, rather than live experimentation. This ensures that learning occurs within the safe confines of a controlled simulation environment and not at all at the level of patient trials.

From an ethical perspective, all deployments must be done according to the accepted principles of medical ethics: beneficence, non-maleficence, autonomy, and justice. The algorithmic decision-making should also be transparent; clinicians must be able to interpret how dosage recommendations are made. Future real-world applications of such systems should also include human-in-the-loop supervision, where ultimate decision authority rests with the medical professional over the AI-suggested dosage.

In order for fairness and inclusiveness to be ensured in the medical dosages that are determined by the AI systems, the data sets that are used for training the systems should include diverse populations of patients in terms of age groups, gender, genetic backgrounds, and co-existing diseases, etc., with no biases

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